

COMPLETE GYM & HOME WORKOUT GUIDE

12 Muscle Groups • 60+ Exercises • Beginner Friendly

GYM WORKOUTS

Equipment-based training

HOME WORKOUTS

No equipment needed

12

Muscle Groups

60+

Exercises

5+

Per Group

2

Categories

Designed for beginners • AI Gym & Fitness Assistant Project

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This guide covers **12 major muscle groups** with **5 gym-based** and **5 home-based** exercises each. Every exercise includes the exact primary and secondary muscles involved, total muscle count, sets/reps, and a beginner tip. Gym exercises require equipment; home exercises use bodyweight, household items, or minimal resistance.

PART 1: GYM WORKOUTS

Equipment-based training — barbell, dumbbells, cables & machines

The following exercises are designed for a gym environment. They use barbells, dumbbells, cable machines, and resistance equipment to provide progressive overload — the key driver of muscle growth and strength gains.

■ Chest (Pectorals) — Gym Exercises

The pectoral muscles form the chest wall. They are primarily responsible for arm adduction, flexion, and medial rotation. A strong chest improves posture and upper-body power.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Barbell Bench Press	4 sets × 8–10 reps	<i>Pectoralis Major (sternocostal head)</i>	<i>Anterior Deltoid, Triceps Brachii</i>	4	■ Keep shoulder blades retracted and feet flat on the floor throughout.
Incline Dumbbell Press	3 sets × 10–12 reps	<i>Pectoralis Major (clavicular head)</i>	<i>Anterior Deltoid, Triceps Brachii</i>	4	■ Set bench to 30–45°. Higher angles shift work to the shoulders.
Cable Crossover (Chest Fly)	3 sets × 12–15 reps	<i>Pectoralis Major, Pectoralis Minor</i>	<i>Anterior Deltoid, Serratus Anterior</i>	5	■ Squeeze the chest at the midpoint — pause for 1 second.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Machine Chest Press	3 sets × 10–12 reps	<i>Pectoralis Major</i>	<i>Triceps Brachii, Anterior Deltoid</i>	4	■ Great for beginners — the fixed path reduces injury risk.
Dips (chest-forward lean)	3 sets × 8–12 reps	<i>Pectoralis Major (lower), Pectoralis Minor</i>	<i>Triceps Brachii, Anterior Deltoid</i>	5	■ Lean forward 30° to maximize chest activation over triceps.

■ Back (Latissimus Dorsi & Trapezius) — Gym Exercises

The back contains the body's largest upper-body muscles. The lats give the V-taper shape; the traps support posture and neck. A strong back prevents injuries from everyday activities.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Barbell Deadlift	4 sets × 5–6 reps	<i>Latissimus Dorsi, Erector Spinae</i>	<i>Gluteus Maximus, Hamstrings, Trapezius, Rhomboids</i>	8	■ Hip hinge first — push the floor away, don't pull the bar up.
Lat Pulldown	4 sets × 10–12 reps	<i>Latissimus Dorsi</i>	<i>Biceps Brachii, Rear Deltoid, Rhomboids, Teres Major</i>	6	■ Pull to the upper chest; avoid swinging the torso back.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Seated Cable Row	3 sets × 10–12 reps	<i>Latissimus Dorsi, Rhomboids</i>	<i>Biceps Brachii, Trapezius, Rear Deltoid</i>	6	■ Keep the lower back neutral — don't round at the top of the row.
Barbell Bent-Over Row	4 sets × 8–10 reps	<i>Latissimus Dorsi, Rhomboids, Trapezius</i>	<i>Biceps Brachii, Erector Spinae, Posterior Deltoid</i>	7	■ Torso at 45°, bar pulled to the lower belly for maximum lat stretch.
Single-Arm Dumbbell Row	3 sets × 10–12 reps per side	<i>Latissimus Dorsi, Rhomboids</i>	<i>Biceps Brachii, Trapezius, Core Stabilizers</i>	6	■ Let the shoulder blade move freely — full range of motion matters.

■ Shoulders (Deltoids) — Gym Exercises

The deltoid has three heads: anterior (front), medial (side), and posterior (rear). Together they allow 360° arm movement. Well-developed shoulders create the appearance of a broader physique.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Overhead Barbell Press (OHP)	4 sets × 6–8 reps	<i>Anterior Deltoid, Medial Deltoid</i>	<i>Triceps Brachii, Trapezius, Core Stabilizers</i>	6	■ Press the bar in a slight backward arc; don't lock elbows aggressively.
Dumbbell Lateral Raises	4 sets × 12–15 reps	<i>Medial Deltoid</i>	<i>Supraspinatus, Trapezius (upper)</i>	4	■ Lead with the elbows, not the hands. Stop at shoulder height.
Face Pulls (cable)	3 sets × 15–20 reps	<i>Posterior Deltoid, Rotator Cuff (external rotators)</i>	<i>Rhomboids, Trapezius (mid), Infraspinatus</i>	6	■ Pull to forehead level; flare elbows out and squeeze the rear delts.
Arnold Press	3 sets × 10–12 reps	<i>Anterior, Medial & Posterior Deltoid</i>	<i>Triceps Brachii, Trapezius, Rotator Cuff</i>	7	■ Rotate palms inward at the bottom; rotate outward as you press up.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Reverse Pec-Deck (Rear Delt Machine)	3 sets × 12–15 reps	<i>Posterior Deltoid</i>	<i>Rhomboids, Trapezius (mid), Infraspinatus</i>	5	■ <i>Adjust the seat so handles align with shoulders; maintain a slight bend in elbows.</i>

■ Biceps — Gym Exercises

The *biceps brachii* has two heads (long and short) and is the primary elbow flexor. It also assists in forearm supination. Despite being a small muscle, it is one of the most trained due to its visual prominence.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Barbell Bicep Curl	4 sets × 8–10 reps	<i>Biceps Brachii (long & short head)</i>	<i>Brachialis, Brachioradialis</i>	4	■ <i>Keep elbows pinned to sides — don't swing the torso to cheat the weight up.</i>
Incline Dumbbell Curl	3 sets × 10–12 reps	<i>Biceps Brachii (long head, fully stretched)</i>	<i>Brachialis, Anterior Deltoid</i>	4	■ <i>Incline stretches the long head for maximum growth stimulus.</i>

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Cable Hammer Curl	3 sets × 12–15 reps	<i>Brachialis, Brachioradialis</i>	<i>Biceps Brachii, Forearm Flexors</i>	5	Neutral grip (thumbs up) shifts emphasis from bicep to brachialis — adds arm thickness.
Preacher Curl (EZ-bar)	3 sets × 10–12 reps	<i>Biceps Brachii (short head)</i>	<i>Brachialis, Brachioradialis</i>	4	Don't let the bar crash down — control the eccentric (lowering) phase.
Concentration Curl	3 sets × 12 reps per arm	<i>Biceps Brachii (peak contraction)</i>	<i>Brachialis</i>	3	Rest your elbow on inner thigh; supinate wrist at the top for maximum peak.

■ Triceps — Gym Exercises

The triceps brachii has three heads (long, medial, lateral) and makes up roughly 65% of the upper arm's size. It is the primary elbow extensor and is heavily involved in all pressing movements.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Close-Grip Bench Press	4 sets × 8–10 reps	<i>Triceps Brachii (all 3 heads)</i>	<i>Pectoralis Major (inner), Anterior Deltoid</i>	5	■ Hands shoulder-width apart; tuck elbows in at 45° — not flared out.
Skull Crushers (EZ-bar)	3 sets × 10–12 reps	<i>Triceps Brachii (long head)</i>	<i>Anconeus, Posterior Deltoid (stabilizer)</i>	4	■ Lower bar to the forehead (or just behind it) for maximum long-head stretch.
Tricep Rope Pushdown	3 sets × 12–15 reps	<i>Triceps Brachii (lateral & medial heads)</i>	<i>Anconeus, Forearm Extensors</i>	4	■ Spread the rope at the bottom; fully extend and pronate for peak contraction.
Overhead Tricep Extension	3 sets × 10–12 reps	<i>Triceps Brachii (long head — fully stretched)</i>	<i>Anconeus, Core Stabilizers</i>	4	■ Elbows point straight up; don't let them flare outward as you lower the weight.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Tricep Dips (parallel bars)	3 sets × 8–12 reps	<i>Triceps Brachii (all 3 heads)</i>	<i>Pectoralis Major (lower), Anterior Deltoid</i>	5	■ Keep torso upright to target triceps; lean forward to shift work to the chest.

■ Quadriceps — Gym Exercises

The quadriceps group consists of four muscles (*rectus femoris*, *vastus lateralis*, *vastus medialis*, *vastus intermedius*). They are the primary knee extensors and among the most powerful muscles in the body.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Barbell Back Squat	4 sets × 6–8 reps	<i>Quadriceps (all 4 heads), Gluteus Maximus</i>	<i>Hamstrings, Adductors, Erector Spinae, Core</i>	9	■ Depth at or below parallel; knees track over toes throughout the movement.
Leg Press (machine)	4 sets × 10–12 reps	<i>Quadriceps (vastus lateralis, vastus medialis)</i>	<i>Gluteus Maximus, Hamstrings, Adductors</i>	6	■ Feet low and close = more quad; feet high and wide = more glutes and hamstrings.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Leg Extension (machine)	3 sets × 12–15 reps	<i>Quadriceps (rectus femoris, vastus muscles)</i>	<i>Tibialis Anterior (stabilizer)</i>	5	■ Great isolation tool — use lighter weight and slow the eccentric (4 sec down).
Hack Squat (machine)	3 sets × 10–12 reps	<i>Quadriceps (especially vastus medialis — inner quad)</i>	<i>Gluteus Maximus, Hamstrings</i>	5	■ Feet low and narrow for maximum quad emphasis; track knees over toes.
Walking Lunges (barbell or dumbbell)	3 sets × 12 reps per leg	<i>Quadriceps, Gluteus Maximus</i>	<i>Hamstrings, Adductors, Core Stabilizers, Tibialis Anterior</i>	7	■ Take a long step; front knee should not go past the toes as you lunge forward.

■ Hamstrings — Gym Exercises

The hamstring group includes three muscles: *biceps femoris*, *semitendinosus*, and *semimembranosus*. They flex the knee and extend the hip. Often undertrained, leading to injury and muscle imbalance.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Romanian Deadlift (RDL)	4 sets × 8–10 reps	<i>Hamstrings (biceps femoris, semitendinosus)</i>	<i>Gluteus Maximus, Erector Spinae, Adductors</i>	6	■ Hinge at the hips; bar stays close to legs. Feel the hamstring stretch — stop before rounding.
Lying Leg Curl (machine)	3 sets × 10–12 reps	<i>Hamstrings (all 3 muscles)</i>	<i>Gastrocnemius (secondary knee flexor)</i>	4	■ Control the eccentric — 3 seconds down. Don't let hips rise off the pad.
Seated Leg Curl (machine)	3 sets × 12–15 reps	<i>Hamstrings (especially biceps femoris short head)</i>	<i>Gastrocnemius, Popliteus</i>	4	■ Knees at 90° or slightly more at start — seated position stretches hamstrings differently.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Good Mornings	3 sets × 10 reps	<i>Hamstrings, Erector Spinae</i>	<i>Gluteus Maximus, Adductors, Core</i>	6	■ Hinge forward with bar on traps; only go as low as flexibility allows without rounding.
Sumo Deadlift	3 sets × 6–8 reps	<i>Hamstrings, Gluteus Maximus, Adductors</i>	<i>Quadriceps, Erector Spinae, Trapezius</i>	7	■ Wide stance, toes flared out 45°. Push the knees out and pull the floor apart.

■ Glutes — Gym Exercises

The gluteal group includes *gluteus maximus* (largest muscle in the body), *gluteus medius*, and *gluteus minimus*. They extend, abduct, and rotate the hip. Critical for athletic performance and lower-back health.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Barbell Hip Thrust	4 sets × 10–12 reps	<i>Gluteus Maximus</i>	<i>Hamstrings, Adductors, Core</i>	5	■ Bar padded on hips; squeeze glutes hard at the top — hold for 2 seconds.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Cable Kickback	3 sets × 15 reps per leg	<i>Gluteus Maximus</i>	<i>Hamstrings, Core Stabilizers</i>	4	■ Kick straight back — not to the side. Squeeze the glute at full extension.
Smith Machine Sumo Squat	3 sets × 12 reps	<i>Gluteus Maximus, Gluteus Medius</i>	<i>Adductors, Quadriceps, Hamstrings</i>	6	■ Wide stance, toes out 45°. The Smith machine allows you to sit back safely.
Abductor Machine	3 sets × 15–20 reps	<i>Gluteus Medius, Gluteus Minimus</i>	<i>Tensor Fasciae Latae, Piriformis</i>	5	■ Push knees apart against resistance; hold for 1 sec at the top of each rep.
Romanian Deadlift (glute focus)	4 sets × 10 reps	<i>Gluteus Maximus, Hamstrings</i>	<i>Erector Spinae, Adductors, Core</i>	6	■ Squeeze glutes as you return to standing — imagine crushing a walnut between them.

■ Core (Abs & Obliques) — Gym Exercises

The core includes the rectus abdominis (6-pack), internal and external obliques, transverse abdominis (deepest layer), and erector spinae. A strong core stabilizes the spine in every movement.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Cable Crunch	4 sets × 15–20 reps	Rectus Abdominis	External Obliques, Internal Obliques	4	■ Pull the elbows to the knees; the movement comes from the abs — not the hips or arms.
Hanging Leg Raise	3 sets × 12–15 reps	Rectus Abdominis (lower), Hip Flexors	Obliques, Transverse Abdominis, Serratus Anterior	6	■ Control the descent — no swinging. For beginners, bend the knees to reduce difficulty.
Ab Rollout (ab wheel)	3 sets × 8–12 reps	Rectus Abdominis, Transverse Abdominis	Obliques, Erector Spinae, Latissimus Dorsi	6	■ Keep hips low and core braced throughout; don't let the lower back sag.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Russian Twist (with plate or cable)	3 sets × 20 reps (each side)	<i>External & Internal Obliques</i>	<i>Rectus Abdominis, Hip Flexors, Erector Spinae</i>	6	■ Feet off the floor for more difficulty; rotate from the torso — not just the arms.
Decline Crunch	3 sets × 15–20 reps	<i>Rectus Abdominis</i>	<i>External Obliques, Hip Flexors</i>	4	■ Decline adds resistance via gravity. Hold a plate on the chest for more difficulty.

■ Calves (Gastrocnemius & Soleus) — Gym Exercises

The calf has two main muscles: the gastrocnemius (visible, gives the diamond shape) and the soleus (deeper, endurance-focused). They plantarflex the foot and are used in every step. Notoriously difficult to grow.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Standing Calf Raise (machine or barbell)	4 sets × 15–20 reps	<i>Gastrocnemius (both heads)</i>	<i>Soleus, Peroneus Longus</i>	4	■ Full range — all the way up on the toes, all the way down for a deep stretch. Pause at the bottom.
Seated Calf Raise (machine)	4 sets × 15–20 reps	<i>Soleus (primary when knee is bent)</i>	<i>Gastrocnemius (stretched position), Flexor Hallucis Longus</i>	4	■ Bent knee removes gastrocnemius — isolates the soleus for thickness and endurance.
Donkey Calf Raise	3 sets × 15–20 reps	<i>Gastrocnemius (maximum stretch at bent-hip angle)</i>	<i>Soleus, Peroneus Longus</i>	4	■ Hips at 90°, weight on the lower back. Old-school but extremely effective.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Leg Press Calf Raise	3 sets × 20 reps	<i>Gastrocnemius, Soleus</i>	<i>Tibialis Anterior (antagonist), Peroneus Longus</i>	5	■ Press the toe plate with the balls of the feet — avoid using the toes alone.
Single-Leg Calf Raise (with dumbbell)	3 sets × 12–15 reps per leg	<i>Gastrocnemius (single-leg loading)</i>	<i>Soleus, Peroneus Longus, Balance Muscles (Peroneals)</i>	5	■ One hand holds a dumbbell; press high onto the ball of the foot for full range.

■ Forearms & Grip — Gym Exercises

The forearms contain over 20 muscles divided into flexor and extensor compartments. They control wrist and finger movement and are essential for grip strength in all pulling exercises.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Wrist Roller	3 sets × 3 rolls (up and down)	<i>Flexor Carpi Radialis, Flexor Carpi Ulnaris</i>	<i>Extensor Carpi Radialis, Brachioradialis, Finger Flexors</i>	6	■ Arms extended forward; wind the weight up and unwind it slowly. Brutal but effective.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Reverse Barbell Curl	3 sets × 10–12 reps	<i>Brachioradialis, Extensor Carpi Radialis</i>	<i>Biceps Brachii, Flexor Carpi Radialis</i>	5	■ Pronated grip (palms down); keep elbows stable at your sides.
Barbell Wrist Curl	3 sets × 15–20 reps	<i>Flexor Carpi Radialis, Flexor Carpi Ulnaris, Palmaris Longus</i>	<i>Finger Flexors (FDS, FDP)</i>	6	■ Forearms resting on bench; curl the wrists upward and fully lower for the stretch.
Farmer's Carry	3 sets × 30–40 metres	<i>Finger Flexors, Intrinsic Hand Muscles</i>	<i>Trapezius, Core Stabilizers, Erector Spinae, Deltoids</i>	8	■ Heavy dumbbells or plates; walk with purpose. Builds real-world grip strength fast.
Dead Hang (from pull-up bar)	3 sets × 30–60 sec	<i>Finger Flexors, Intrinsic Hand Muscles</i>	<i>Latissimus Dorsi, Trapezius, Brachioradialis</i>	7	■ Hang from the bar with straight arms — decompresses the spine and builds grip endurance.

■ Lower Back (Erector Spinae) — Gym Exercises

The erector spinae runs the length of the spine and consists of three columns (*iliocostalis*, *longissimus*, *spinalis*). Together with the multifidus, they maintain upright posture and allow spinal extension. Critical to train safely.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Hyperextensions (back extension machine)	3 sets × 12–15 reps	<i>Erector Spinae (all 3 columns)</i>	<i>Gluteus Maximus, Hamstrings, Multifidus</i>	5	■ Round down fully at the bottom; extend only to neutral — don't hyperextend at the top.
Deadlift (conventional)	4 sets × 5 reps	<i>Erector Spinae, Gluteus Maximus</i>	<i>Hamstrings, Latissimus Dorsi, Trapezius, Core</i>	8	■ Spine neutral from setup to lockout. This is the most important rule in the deadlift.
Good Mornings (barbell)	3 sets × 8–10 reps	<i>Erector Spinae, Hamstrings</i>	<i>Gluteus Maximus, Multifidus, Core</i>	6	■ Light weight; hinge at the hip. Stop when you feel mild hamstring tension.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Seated Cable Row (back focus)	3 sets × 10–12 reps	<i>Erector Spinae (stabilizer), Latissimus Dorsi</i>	<i>Rhomboids, Trapezius, Biceps</i>	6	■ Sit tall and avoid excessive torso sway — the lower back works isometrically.
Rack Pull (partial deadlift)	3 sets × 5–6 reps	<i>Erector Spinae, Trapezius</i>	<i>Gluteus Maximus, Hamstrings, Latissimus Dorsi</i>	6	■ Bar starts at knee height — allows heavier loads and safe lower back loading for beginners.

PART 2: HOME WORKOUTS

No gym needed — bodyweight, household items & minimal equipment

These exercises can be done at home with zero or minimal equipment. Bodyweight training is surprisingly effective — many elite athletes build strength entirely without a gym. Master these before, or alongside, gym training.

■ Chest (Pectorals) — Home Exercises

The pectoral muscles form the chest wall. They are primarily responsible for arm adduction, flexion, and medial rotation. A strong chest improves posture and upper-body power.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Standard Push-Ups	4 sets × 15–20 reps	<i>Pectoralis Major</i>	<i>Anterior Deltoid, Triceps Brachii, Core Stabilizers</i>	5	■ <i>Body forms a straight line from head to heels the whole time.</i>
Wide-Grip Push-Ups	3 sets × 12–15 reps	<i>Pectoralis Major (outer fibres)</i>	<i>Anterior Deltoid, Triceps Brachii</i>	4	■ <i>Hands placed 6–8 inches wider than shoulder-width.</i>
Decline Push-Ups (feet elevated)	3 sets × 10–12 reps	<i>Pectoralis Major (upper/clavicular head)</i>	<i>Anterior Deltoid, Triceps, Serratus Anterior</i>	5	■ <i>Use a chair or sofa to elevate your feet by 30–45 cm.</i>

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Diamond Push-Ups	3 sets × 10–15 reps	<i>Pectoralis Major (inner), Triceps Brachii</i>	<i>Anterior Deltoid, Core</i>	5	■ Hands form a diamond shape directly under the chest.
Plyometric Push-Ups (Clap Push-Ups)	3 sets × 8–10 reps	<i>Pectoralis Major, Fast-twitch fibres</i>	<i>Triceps, Anterior Deltoid, Core</i>	5	■ Explosive movement builds power. Land softly with elbows slightly bent.

■ Back (Latissimus Dorsi & Trapezius) — Home Exercises

The back contains the body's largest upper-body muscles. The lats give the V-taper shape; the traps support posture and neck. A strong back prevents injuries from everyday activities.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Pull-Ups	4 sets × max reps (aim 5–12)	<i>Latissimus Dorsi</i>	<i>Biceps Brachii, Rhomboids, Teres Major, Core</i>	6	■ Dead hang at the bottom; chin over bar at the top. Full ROM.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Chin-Ups	3 sets × 8–10 reps	<i>Latissimus Dorsi, Biceps Brachii</i>	<i>Rhomboids, Posterior Deltoid, Core</i>	6	■ <i>Supinated grip (palms facing you) increases bicep involvement.</i>
Inverted Rows (using a table)	3 sets × 12–15 reps	<i>Latissimus Dorsi, Rhomboids</i>	<i>Biceps, Trapezius, Core, Posterior Deltoid</i>	6	■ <i>Lie under a sturdy table; pull chest up to edge. Keep body straight.</i>
Superman Hold	3 sets × 15–20 reps (3 sec hold)	<i>Erector Spinae, Trapezius (lower fibres)</i>	<i>Gluteus Maximus, Posterior Deltoid, Rhomboids</i>	5	■ <i>Lift arms and legs simultaneously; squeeze the glutes at the top.</i>
Towel Row (door anchor)	3 sets × 12–15 reps	<i>Latissimus Dorsi, Rhomboids</i>	<i>Biceps, Trapezius, Core Stabilizers</i>	5	■ <i>Loop a towel around a door handle; lean back and row your body forward.</i>

■ Shoulders (Deltoids) — Home Exercises

The deltoid has three heads: anterior (front), medial (side), and posterior (rear). Together they allow 360° arm movement. Well-developed shoulders create the appearance of a broader physique.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Pike Push-Ups	4 sets × 10–15 reps	<i>Anterior Deltoid, Medial Deltoid</i>	<i>Triceps Brachii, Trapezius, Core</i>	6	■ Hips high, body in inverted V. Lower the crown of the head toward the floor.
Handstand Push-Ups (wall-assisted)	3 sets × 5–8 reps	<i>Anterior, Medial & Posterior Deltoid</i>	<i>Triceps, Trapezius, Core, Serratus Anterior</i>	7	■ Kick up against a wall, lower slowly to the floor, press back up explosively.
Dumbbell / Water-Bottle Side Raises	3 sets × 15–20 reps	<i>Medial Deltoid</i>	<i>Supraspinatus, Trapezius (upper)</i>	4	■ Use filled water bottles (1–2 L each) as resistance if no dumbbells are available.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Prone Y-T-W Raises	3 sets × 10 reps each position	<i>Posterior Deltoid, Trapezius (lower and mid)</i>	<i>Rhomboids, Infraspinatus, Rotator Cuff</i>	6	■ Lie face-down; form Y, T and W shapes with arms. Excellent for posture correction.
Arm Circles	3 sets × 30 seconds each direction	<i>Medial Deltoid, Anterior Deltoid</i>	<i>Rotator Cuff (all 4 muscles), Trapezius</i>	7	■ Start small circles and gradually widen. Great warm-up and rehab tool.

■ Biceps — Home Exercises

The biceps brachii has two heads (long and short) and is the primary elbow flexor. It also assists in forearm supination. Despite being a small muscle, it is one of the most trained due to its visual prominence.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Chin-Ups (supinated grip)	4 sets × 6–10 reps	<i>Biceps Brachii, Latissimus Dorsi</i>	<i>Brachialis, Rhomboids, Core</i>	6	■ Palms facing you — this doubles bicep activation compared to pull-ups.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Towel / Bedsheet Curl	3 sets × 12–15 reps	<i>Biceps Brachii</i>	<i>Brachialis, Forearm Flexors</i>	4	■ Loop a towel under your foot; grip both ends and curl your arms upward.
Backpack Curl	3 sets × 10–15 reps	<i>Biceps Brachii</i>	<i>Brachialis, Brachioradialis</i>	4	■ Fill a backpack with books for resistance. Control both up and down phases.
Isometric Bicep Hold	3 sets × 30–45 sec	<i>Biceps Brachii (static contraction)</i>	<i>Brachialis, Forearm Flexors</i>	4	■ Hold a heavy bag at 90° elbow flexion — creates intense time-under-tension.
Resistance Band Curl	4 sets × 15–20 reps	<i>Biceps Brachii</i>	<i>Brachialis, Brachioradialis</i>	4	■ Stand on the band; in increasing tension at the top trains the contracted position harder.

■ Triceps — Home Exercises

The triceps brachii has three heads (long, medial, lateral) and makes up roughly 65% of the upper arm's size. It is the primary elbow extensor and is heavily involved in all pressing movements.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Diamond Push-Ups	4 sets × 10–15 reps	<i>Triceps Brachii (all heads)</i>	<i>Pectoralis Major (inner), Anterior Deltoid</i>	5	■ Hands touching in a diamond shape under the sternum; elbows brush your sides.
Chair / Bench Dips	3 sets × 12–15 reps	<i>Triceps Brachii</i>	<i>Anterior Deltoid, Pectoralis Major (lower)</i>	4	■ Legs straight = harder; legs bent = easier. Keep hips close to the chair edge.
Pike Push-Up (tricep focus)	3 sets × 10–12 reps	<i>Triceps Brachii, Anterior Deltoid</i>	<i>Medial Deltoid, Core</i>	5	■ Narrow hand placement shifts load away from shoulders and toward triceps.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Tricep Push-Up (elbows-in)	3 sets × 12–15 reps	<i>Triceps Brachii (medial & lateral head)</i>	<i>Pectoralis Major, Core Stabilizers</i>	5	■ <i>Standard push-up with hands directly under the shoulders; elbows glide backward.</i>
Overhead Towel Extension	3 sets × 12–15 reps	<i>Triceps Brachii (long head)</i>	<i>Core Stabilizers, Posterior Deltoid</i>	4	■ <i>Hold a towel overhead; anchor the top end with one hand, curl the other arm down.</i>

■ Quadriceps — Home Exercises

The quadriceps group consists of four muscles (*rectus femoris*, *vastus lateralis*, *vastus medialis*, *vastus intermedius*). They are the primary knee extensors and among the most powerful muscles in the body.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Bodyweight Squats	4 sets × 20–25 reps	<i>Quadriceps (all 4 heads)</i>	<i>Gluteus Maximus, Hamstrings, Core, Adductors</i>	6	■ <i>Arms forward for balance; sit back like sitting in a chair. Go below parallel.</i>

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Jump Squats	3 sets × 15 reps	<i>Quadriceps, Gluteus Maximus</i>	<i>Hamstrings, Calves, Core</i>	6	■ Explode up maximally; land softly with bent knees to protect the joints.
Bulgarian Split Squat (rear foot elevated)	3 sets × 10–12 reps per leg	<i>Quadriceps (rectus femoris), Gluteus Maximus</i>	<i>Hamstrings, Adductors, Core</i>	6	■ Rear foot on a chair or sofa; 80% of weight on the front leg.
Wall Sit (isometric)	3 sets × 45–60 sec	<i>Quadriceps (all heads — static contraction)</i>	<i>Gluteus Maximus, Core Stabilizers</i>	5	■ Thighs parallel to the floor; back flat against the wall. No sliding down!
Step-Ups (using stairs or a chair)	3 sets × 15 reps per leg	<i>Quadriceps, Gluteus Maximus</i>	<i>Hamstrings, Adductors, Core</i>	6	■ Drive through the heel of the leading foot; don't push off the trailing leg.

■ Hamstrings — Home Exercises

The hamstring group includes three muscles: biceps femoris, semitendinosus, and semimembranosus. They flex the knee and extend the hip. Often undertrained, leading to injury and muscle imbalance.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Nordic Hamstring Curl	3 sets × 5–8 reps	Hamstrings (biceps femoris, semitendinosus)	Gastrocnemius, Gluteus Maximus	5	■ Anchor feet under sofa; lower body slowly forward using hamstrings as a brake.
Glute-Ham Bridge (single leg)	3 sets × 12–15 reps per leg	Hamstrings, Gluteus Maximus	Core Stabilizers, Erector Spinae	5	■ One leg extended; drive hips up through the heel. Hold for 1 sec at the top.
Good Mornings (bodyweight)	3 sets × 15 reps	Hamstrings, Erector Spinae	Gluteus Maximus, Core Stabilizers	5	■ Hands on hips or behind head; hinge forward at hip, soft bend in knee.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Prone Hamstring Curl (ankle weights or resistance band)	3 sets × 15 reps per leg	<i>Hamstrings (all 3 muscles)</i>	<i>Gluteus Maximus (stabilizer), Core</i>	4	■ Lie face-down ; bend the knee, bringing the heel toward the glutes. Squeeze at the top.
Inchworm Exercise	3 sets × 10 reps	<i>Hamstrings (eccentric stretch), Core</i>	<i>Erector Spinae, Anterior Deltoid, Gluteus Maximus</i>	6	■ Stand, hinge forward to floor, walk hands out to a plank, walk backup. Dynamic stretch + strength.

■ Glutes — Home Exercises

The gluteal group includes *gluteus maximus* (largest muscle in the body), *gluteus medius*, and *gluteus minimus*. They extend, abduct, and rotate the hip. Critical for athletic performance and lower-back health.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Glute Bridge	4 sets × 20 reps	<i>Gluteus Maximus</i>	<i>Hamstrings, Core, Erector Spinae</i>	5	■ Drive through heels; squeeze at the top and hold for 2 seconds. Don't hyperextend the lower back.
Single-Leg Glute Bridge	3 sets × 15 reps per leg	<i>Gluteus Maximus</i>	<i>Hamstrings, Core Stabilizers, Erector Spinae</i>	5	■ One leg straight in the air; hold at the top. Prevents left-right imbalances.
Clamshells (with band if available)	3 sets × 20 reps per side	<i>Gluteus Medius, Gluteus Minimus</i>	<i>Piriformis, Tensor Fasciae Latae</i>	5	■ Lie on side, knees bent to 90°; open the top knee like a clamshell while feet stay together.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Donkey Kicks	3 sets × 15–20 reps per leg	<i>Gluteus Maximus</i>	<i>Core Stabilizers, Erector Spinae</i>	4	■ On all fours; kick one leg straight back and up. Keep hips level — don't rotate.
Jump Squats	3 sets × 15 reps	<i>Gluteus Maximus, Quadriceps</i>	<i>Hamstrings, Calves, Core</i>	6	■ Land softly; immediately drop into the next squat. Focus on driving the hips back and down.

■ Core (Abs & Obliques) — Home Exercises

The core includes the *rectus abdominis* (6-pack), *internal and external obliques*, *transverse abdominis* (deepest layer), and *erector spinae*. A strong core stabilizes the spine in every movement.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Plank	3 sets × 45–90 sec	<i>Transverse Abdominis, Rectus Abdominis</i>	<i>Obliques, Erector Spinae, Gluteus Maximus, Anterior Deltoid</i>	7	■ <i>Straight line from head to heels; squeeze glutes and abs together. No sagging hips!</i>
Crunches	3 sets × 25–30 reps	<i>Rectus Abdominis (upper)</i>	<i>External Obliques, Hip Flexors</i>	4	■ <i>Only lift the shoulder blades — your lower back stays on the floor the whole time.</i>
Bicycle Crunches	3 sets × 20 reps each side	<i>Rectus Abdominis, External & Internal Obliques</i>	<i>Hip Flexors, Transverse Abdominis</i>	6	■ <i>Slow and controlled — the most effective ab exercise per EMG research.</i>

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Leg Raises (lying)	3 sets × 15–20 reps	<i>Rectus Abdominis (lower), Hip Flexors</i>	<i>Transverse Abdominis, Obliques</i>	5	■ Press the lower back into the floor throughout ; lower legs just above the ground.
Side Plank	3 sets × 30–60 sec per side	<i>External & Internal Obliques, Quadratus Lumborum</i>	<i>Transverse Abdominis, Gluteus Medius, Anterior Deltoid</i>	7	■ Stack feet or stagger them; raise the top arm vertically for balance challenge.

■ Calves (Gastrocnemius & Soleus) — Home Exercises

The calf has two main muscles: the gastrocnemius (visible, gives the diamond shape) and the soleus (deeper, endurance-focused). They plantarflex the foot and are used in every step. Notoriously difficult to grow.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Bodyweight Standing Calf Raise	4 sets × 25–30 reps	<i>Gastrocnemius</i>	<i>Soleus, Peroneus Longus</i>	4	■ Use a step for deeper range. Go slow: 2 sec up, 1 sec hold at top, 3 sec down.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Single-Leg Bodyweight Calf Raise	3 sets × 20 reps per leg	<i>Gastrocnemius (single-limb overload)</i>	<i>Soleus, Peroneals, Tibialis Posterior</i>	5	■ Hold a wall for balance; elevate the opposite foot behind you. Add a loaded backpack for resistance.
Jump Rope (skipping)	3 sets × 2 minutes	<i>Gastrocnemius, Soleus (repetitive plantar flexion)</i>	<i>Tibialis Anterior, Peroneus Longus, Quadriceps</i>	6	■ Stay on the balls of the feet; minimal ground contact time for maximum calf activation.
Explosive Calf Jumps	3 sets × 20 reps	<i>Gastrocnemius (power fibres)</i>	<i>Soleus, Achilles Tendon (elastic recoil), Quadriceps</i>	5	■ Hop with straight legs using only the calves — like a pogo stick. Short, rapid hops.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Seated Towel Calf Raise	3 sets × 20 reps	<i>Soleus (bent-knee isolates soleus)</i>	<i>Gastrocnemius, Flexor Hallucis Longus</i>	4	■ Sit on a chair, place a heavy bag on your thighs; push onto tiptoes and lower slowly.

■ Forearms & Grip — Home Exercises

The forearms contain over 20 muscles divided into flexor and extensor compartments. They control wrist and finger movement and are essential for grip strength in all pulling exercises.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Towel Wringing	3 sets × 30 sec each direction	<i>Finger Flexors, Forearm Flexors & Extensors</i>	<i>Intrinsic Hand Muscles, Wrist Stabilizers</i>	5	■ Wring a wet towel as tightly as possible in both directions. Simple but remarkably effective.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Rice Bucket Training	3 sets × 2 minutes	<i>Intrinsic Hand Muscles, Finger Flexors & Extensors</i>	<i>Forearm Flexors, Forearm Extensors</i>	6	■ Dig, grip, squeeze and rotate the hand in a bucket of rice — used by martial artists and athletes.
Book Pinch Grip	3 sets × 30–45 sec per hand	<i>Thenar & Hypothenar Muscles, Intrinsic Hand Muscles</i>	<i>Flexor Digitorum Superficialis, Flexor Pollicis Longus</i>	6	■ Pinch a large book between thumb and fingers; hold at arm's length. Progress to thicker books.
Resistance Band Wrist Extension	3 sets × 20 reps per hand	<i>Extensor Carpi Radialis, Extensor Carpi Ulnaris</i>	<i>Finger Extensors, Brachioradialis</i>	5	■ Anchor a light band under the foot; with palm facing down, curl the wrist upward.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Reverse Push-Ups (on knuckles)	3 sets × 10–15 reps	<i>Forearm Extensors, Wrist Stabilizers</i>	<i>Pectoralis Major, Triceps, Core</i>	6	■ Perform push-ups on closed fists; strengthens the wrists and builds forearm density.

■ Lower Back (Erector Spinae) — Home Exercises

The erector spinae runs the length of the spine and consists of three columns (*iliocostalis*, *longissimus*, *spinalis*). Together with the multifidus, they maintain upright posture and allow spinal extension. Critical to train safely.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Superman Holds	3 sets × 15 reps (3 sec hold)	<i>Erector Spinae, Multifidus</i>	<i>Gluteus Maximus, Posterior Deltoid, Hamstrings</i>	6	■ Lift arms and legs simultaneously off the floor; squeeze the glutes and hold for 3 seconds.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Bird-Dog	3 sets × 12 reps each side	<i>Erector Spinae, Multifidus, Transverse Abdominis</i>	<i>Gluteus Maximus, Core Stabilizers</i>	6	■ On all fours; extend opposite arm and leg simultaneously. Hold 2 sec. Don't rotate the hips.
Glute Bridge (lower back version)	3 sets × 20 reps	<i>Gluteus Maximus, Erector Spinae (stabilizer)</i>	<i>Hamstrings, Core, Multifidus</i>	6	■ Press through heels; use the glutes to drive up — this protects the lower back from overuse.
Cat-Cow Stretch (active mobility)	3 sets × 15 reps	<i>Erector Spinae, Multifidus (dynamic contraction)</i>	<i>Transverse Abdominis, Rectus Abdominis</i>	5	■ On all fours; arch the back (cow), then round it (cat). Breathe in on cow, out on cat.

Exercise	Sets & Reps	Primary Muscles	Secondary Muscles	Muscles Involved	Beginner Tip
Reverse Snow Angel	3 sets × 15 reps	<i>Erector Spinae, Trapezius (lower), Rhomboids</i>	<i>Posterior Deltoid, Gluteus Maximus</i>	6	■ Lie face-down, arms at sides; sweep arms up overhead and back like a snow angel motion.

Quick Reference Summary

Muscle Group	Primary Muscles Targeted	Gym Exercises	Home Exercises	Max Muscles Involved
Chest (Pectorals)	<i>Pectoralis Major (sternocostal head)</i>	5	5	5
Back (Latissimus Dorsi & Trapezius)	<i>Latissimus Dorsi, Erector Spinae</i>	5	5	8
Shoulders (Deltoids)	<i>Anterior Deltoid, Medial Deltoid</i>	5	5	7
Biceps	<i>Biceps Brachii (long & short head)</i>	5	5	6
Triceps	<i>Triceps Brachii (all 3 heads)</i>	5	5	5
Quadriceps	<i>Quadriceps (all 4 heads), Gluteus Maximus</i>	5	5	9
Hamstrings	<i>Hamstrings (biceps femoris, semitendinosus)</i>	5	5	7
Glutes	<i>Gluteus Maximus</i>	5	5	6
Core (Abs & Obliques)	<i>Rectus Abdominis</i>	5	5	7
Calves (Gastrocnemius & Soleus)	<i>Gastrocnemius (both heads)</i>	5	5	6
Forearms & Grip	<i>Flexor Carpi Radialis, Flexor Carpi Ulnaris</i>	5	5	8
Lower Back (Erector Spinae)	<i>Erector Spinae (all 3 columns)</i>	5	5	8

Beginner's Notes

- **Progressive overload:** Add a small amount of weight or 1–2 reps each week. This is the #1 rule of muscle growth.
- **Recovery:** Muscles grow during rest, not during the workout. Aim for 7–9 hours of sleep and at least 1 rest day between training the same muscle group.
- **Nutrition:** Consume 1.6–2.0 g of protein per kg of bodyweight daily. Calorie surplus to build muscle; calorie deficit to lose fat.
- **Form first:** Never sacrifice form for weight. Injuries set you back months. Learn the movement pattern with light weight first.
- **Consistency:** 3–4 days of training per week for 6+ months is where most beginners see transformative results. Don't rush it.